

CMPE2150 Lab 02

Circuit Isolation

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Optoisolated Level Shifting with a Phototransistor

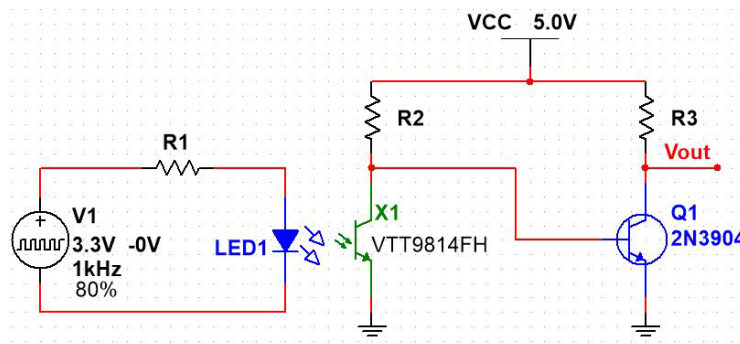


Figure 1: Phototransistor Isolation

For this circuit, note the following:

- V1 is a signal generator (preferably the Analog Discovery 2 without the BNC Breakout board – see next part of this exercise) set to produce a pulse with the following characteristics:
 - Low voltage is 0V, High voltage is 3.3V
 - Frequency is 1kHz
 - Duty cycle is 80%
- The LED is Yellow (check the [spec sheet for the phototransistor](#) to see why) with a forward drop of 1.8V

- The phototransistor, VTT9814FH, is in your CNT Year 2 Kit. It has the Collector marked by a flat spot in the plastic case; again, check the spec sheet to verify this.
- The 2N3904 has a minimum beta of 100, and an “on” V_{BE} of 0.7V
 1. Determine a standard 10% resistor value for R_1 that will provide approximately 10mA of current to the LED: _____ Ω .
 2. Determine a standard 10% resistor value for R_3 that will draw approximately 10mA (go for slightly less current if necessary) through Q_1 when it is saturated. Assume that V_{CEsat} is essentially 0V: _____ Ω .
 3. When X_1 is off, R_2 simply becomes the Base resistor for Q_1 . Determine the maximum value for this resistor required to saturate the transistor, based upon a collector current of 10mA, a V_{BE} of 0.7V, and a beta of 100 for Q_1 . Now, choose a standard 10% resistor value that’s approximately half of your calculated value: _____ $k\Omega$.

Build and Test

Build the circuit. You will need to physically arrange the LED and the photo transistor so they practically touch “top to top” – in other words, the light from the LED must be closely coupled into the photo transistor. To do this, you will have to bend the leads of both of them 90°.

To maintain proper isolation, you should use your Analog Discovery 2 with the BNC breakout board removed so you can use the fly-wires directly. (The BNC breakout board provides common grounds for the scope channels and waveform generators, whereas the fly-wires allow the scope channels to be self-referenced. For each channel, you will need to use the “+” wire for the signal and the “-” wire as the reference – “GND” won’t work, since it’s not the channel reference in this mode of operation.) If you don’t have access to an AD2, you can use bench equipment, you just won’t have the two sides of the circuit completely isolated.

For a grade out of three marks, set up your oscilloscope to display your input and output and ask your instructor to grade your work in class.

_____/3

Remove the signal generator, the oscilloscope, and the DC power supply. Using a DMM as an ohmmeter, check if the resistance between the two halves of your circuit is practically infinite for any of the points in the two circuits: _____ (True or False)

Electrostatic Isolation

Consider the following schematic diagram to answer the questions that follow.

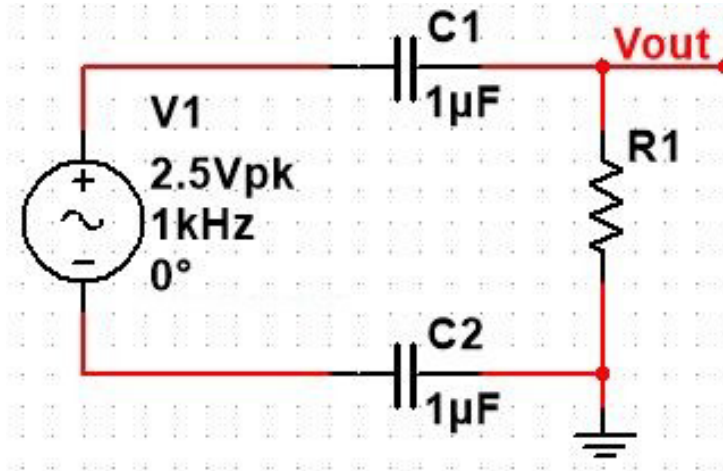


Figure 2: Electrostatic Isolation

- V1 is a signal generator set to produce the following:
 - a Sine Wave
 - with an amplitude of $2.5V_p$
 - and a frequency of $1kHz$
 - with no DC offset
- The two capacitors are effectively in series

The two capacitors and the resistor form a High Pass Filter (HPF) at V_{out} . Determine the resistance which would provide a cutoff frequency one decade below the signal generator frequency, and pick the closest standard 10% resistor value: _____ $k\Omega$.

Build and Test

Build this circuit, again preferably using the Analog Discovery 2 without the BNC breakout board for both the function generator and the oscilloscope.

For a grade out of three marks, set up your oscilloscope to display your input and output and ask your instructor to grade your work in class.

_____/3

Remove the signal generator, the oscilloscope, and the DC power supply. Using a DMM as an ohmmeter, check if the resistance between the two halves of your circuit is practically infinite for any of the points in the two circuits: _____ (True or False)